COURSE SLIDES · WSQ

DevOps Foundation Training



WSQ Course Code: TGS-2024049184

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v4.0 · 20 August 2026

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ADMI
N

COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

AM, PM and Assessment

Take every required SSG digital attendance for this two-day WSQ course.

2

Trainer displays the QR

Use your mobile camera to scan the current session's SSG QR code.

3

Submit successfully

Wait for confirmation before closing the attendance page.

4

75% minimum

Attendance and competent assessment outcomes determine course completion eligibility.

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About the Trainer



Your Trainer

General trainer profile template

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

INDUSTRY EXPERIENCE

CONTACT

personal information is optional

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Founder & Chief Training Officer
Tertiary Infotech Academy Pte Ltd

EXPERIENCE

More than 20 years across software engineering, AI, data analytics and cloud computing.

QUALIFICATIONS

PhD (NUS), DACE, ACTA and GCWLP among professional credentials.

COURSE FOCUS

Agile flow, CI/CD automation, cloud delivery and evidence-led problem management.

TEACHING APPROACH

Connects concepts to workplace decisions, reproducible data and auditable evidence.

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ICE-BREAKER

Let's Know Each Other

1

Introduce

Share your name, organisation and role.

2

Recall

Name one delivery or operational problem you have seen recur.

3

Aim

Choose one outcome: faster flow, safer change, stronger investigation or clearer reporting.

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Ground Rules

1. **Participate**
   Ask, challenge and contribute evidence.

2. **Respect**
   Critique the system or idea, not the person.

3. **Protect data**
   Use only the synthetic datasets in the labs.

4. **Learn from failure**
   A failed script or hypothesis is feedback.

5. **Be punctual**
   Return on time and complete attendance.

6. **Stay focused**
   One conversation and one shared problem at a time.

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VISUAL DOWNLOAD FLOW · LMS-TMS.TERTIARYINFOTECH.COM

Download Your Course Material

1

Open LMS-TMS

Course portal

2

Sign in

Use your registered email

3

Open this course

DevOps Foundation Training

4

Download

Slides, LG, LP and Labs

OPEN BOOK

Keep the Learner Guide and approved course materials available for the assessment.

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14 HOURS DELIVERY · 2 HOURS ASSESSMENT

Lesson Plan — 2 Days

Day 1

- Day 1 · DevOps foundations, flow, feedback, practices and investigation
  - Exploring DevOps and the problem lifecycle
  - Three Ways, constraints and automated feedback
  - CI/CD, SRE, resilience, DevSecOps, ChatOps and Kanban
  - Agile, Lean, ITSM and evidence-led investigation

Day 2

- Day 2 · Culture, automation, measurement, leadership and assessment
  - Culture, behaviour, maturity and priority policy
  - Automation, architecture, toolchains and platforms
  - DORA metrics, reliability and problem documentation
  - Leadership, capstone, WA and Case Study

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WHAT YOU WILL BE ABLE TO DO

Learning Outcomes

1

LO1

Manage the lifecycle of problems using DevOps principles and technologies.

2

LO2

Perform investigations and develop DevOps methods for problem analysis.

3

LO3

Recommend DevOps solutions and monitor documentation for problem solving.

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Skills Framework Alignment — Knowledge

1. **K1**
   Principles of problem management throughout its lifecycle
2. **K2**
   Relevant tools, processes and technologies for problem identification, investigation, analysis and resolution
3. **K3**
   Problem investigation and diagnosis techniques and methodologies
4. **K4**
   Problem prioritisation and sizing techniques, methodologies and parameters
5. **K5**
   Best practices and industry standards in problem-management documentation

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Skills Framework Alignment — Abilities

1

A1
Manage the lifecycle of a wide range of problems

2

A2
Introduce technologies and processes for automated detection of incidents or problems

3

A3
Investigate and deeply analyse a problem to understand its root causes

4

A4
Develop guidelines for prioritising and categorising problems by severity, frequency and implications

5

A5
Recommend solutions that address root causes and minimise recurrence

6

A6
Monitor documentation and tracking of problems encountered and resolved

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EIGHT DEVOPS FOUNDATION DOMAINS · EIGHT R LABS

Course Outline

1

Topic 01 · Exploring DevOps

Purpose · business value · problem lifecycle · K1 · A1

2

Topic 02 · Core DevOps Principles

Three Ways · constraints · learning · automated feedback · K2 · A2

3

Topic 03 · Key DevOps Practices

CI/CD · testing · SRE · resilience · DevSecOps · ChatOps · Kanban · K2 · A2

4

Topic 04 · Business and Technology Frameworks

Agile · Lean · ITSM · safety culture · learning organisation · K3 · A3

5

Topic 05 · Culture, Behaviours and Operating Models

CALMS · cultural debt · Westrum · maturity · team design · K4 · A4

6

Topic 06 · Automation and DevOps Toolchains

CI/CD · cloud · containers · microservices · IaC · AIOps · platform engineering · K4 · A4

7

Topic 07 · Measurement, Metrics and Reporting

DORA · reliability · flow · business value · problem records · K5 · A6

8

Topic 08 · Sharing, Shadowing and Evolving

Enterprise adoption · leadership · roles · communities · critical success factors · K5 · A5

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HANDS-ON WITHOUT PRODUCTION RISK

R Lab Environment

1

R 4.x

Every lab uses base R only; no paid cloud account or external package is required.

2

Self-contained folders

Each lab includes its script, synthetic data, detailed guide, checklist and expected outputs.

3

Reproducible evidence

Scripts validate inputs and emit a LAB NN PASS marker plus CSV/PNG/Markdown evidence.

4

Connected storyline

Northstar Retail evidence progresses from detection to a capstone problem recommendation.

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BEFORE ASSESSMENT

Briefing for Assessment

1

Individual work

No discussion, recording or sharing of assessment scripts.

2

Open book scope

Use course slides, Learner Guide and approved materials only.

3

Time

WA-SAQ: 60 minutes · Case Study: 60 minutes.

4

Submission

Complete the document provided and upload it to the LMS-TMS.

5

Writing

Use your own words; black or blue pen for hard copy; no correction fluid.

6

Outcome

Competent or Not Yet Competent; reassessment and appeal rights apply.

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WA-SAQ + CASE STUDY

Assessment

1

Written Assessment

Five open-ended questions cover K1–K5.

2

Case Study

Three connected tasks cover A1–A6.

3

Traceability

Every question is taught; every ability is practised in the R labs.

4

Trainer-only keys

Answer keys are withheld from LMS and public GitHub.

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ON ASSESSMENT DAY

Assessment Flow

1. **TRAQOM**
   Complete course feedback
2. **Attendance**
   Scan assessment QR
3. **WA-SAQ**
   5 questions · 60 min
4. **Case Study**
   3 tasks · 60 min
5. **Submit**
   LMS upload and sign-off

EVIDENCE BOUNDARY

Only candidate papers are learner-facing; answer keys remain trainer-only.

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CERTIFICATION CONTEXT · NO EXAM DUMPS

Practice Exam & Certification Readiness

1

Official exam

PeopleCert currently lists 40 multiple-choice questions, 60 minutes, open book and a 65% pass mark.

2

Official study guide

The DevOps Institute v3.4 guide includes the blueprint, terminology and two sample exams.

3

Tertiary practice portal

No matching DevOps Foundation bank was listed on 20 August 2026; do not practise on an unrelated certification.

4

Readiness loop

Study the blueprint, apply the R labs, explain trade-offs aloud, then use only authorised sample questions.

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CONTENT SOURCE ≠ DESIGN SOURCE

Source and Version Authority

1

WSQ authority

Approved Course Proposal v2 and TSC mapping define duration, LO, K/A and WA+CS instruments.

2

Legacy coverage

The v3 deck contributes useful subject coverage only; its layout and incorrect cloud K/A slide were not copied.

3

Certification depth

DevOps Institute DOFD v3.4 supplies the eight-domain blueprint and vocabulary.

4

Current research

PeopleCert and DORA update platform, AI and current five-metric performance context.

5

Secondary sources

User-supplied study pages inform study strategy; no proprietary questions are reproduced.

6

Current release

Version v4.0 · 20 August 2026 · all artifacts regenerated from one source.

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01

TOPIC 01 · LO1 · K1 · A1

Exploring DevOps

Purpose · business value · problem lifecycle

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BLUEPRINT WEIGHT · 5 / 40 exam questions

Key Ideas — Exploring DevOps

1

DevOps is a sociotechnical system

People, process and technology optimise one value stream; a tools-only programme is not DevOps.

2

Shared ownership replaces hand-offs

Development, operations, quality and security own flow and reliability together.

3

Small batches expose problems early

Frequent, reversible changes shorten feedback and reduce the blast radius of failure.

4

Problems become learning assets

Detection, investigation, resolution and prevention are connected through explicit records and feedback.

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TOPIC 01 · K1 · A1

DevOps: What It Is — and Is Not

Dimension	DevOps	Common misconception
Identity	A culture and operating model	A new job title or team
Goal	Fast, safe flow of customer value	Maximum deployment volume
Scope	Idea through operate and learn	Only CI/CD automation
Control	Evidence, guardrails and feedback	Removing governance

WHEN IT MATTERS Use this distinction when diagnosing a transformation that has bought tools but retained siloed incentives.

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TOPIC 01 · K1 · A1

The Problem Lifecycle in a DevOps System

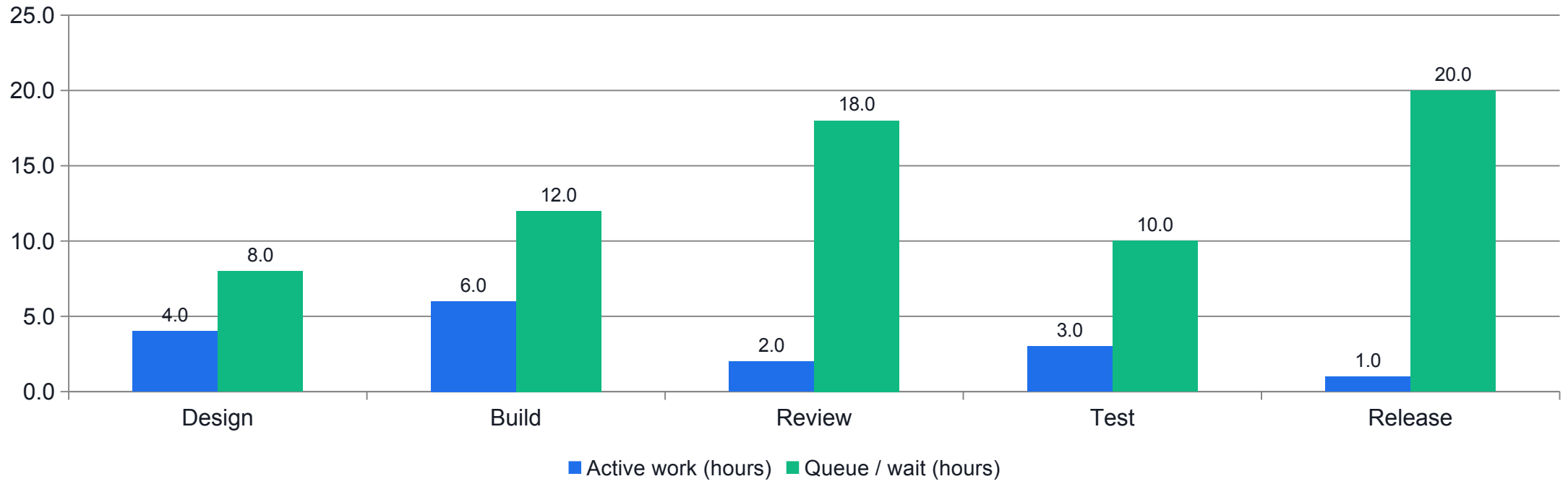


LO1 CONNECTION

A1 is demonstrated by managing the whole lifecycle, not merely closing an incident.

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Why Waiting Dominates Lead Time



WHAT THE DATA SHOWS

The largest improvement opportunity is often the queue between teams, not faster typing inside one stage.

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DevOps Business Outcomes

1

Speed

Shorter lead time turns feedback into value sooner.

2

Stability

Small changes and automation make releases easier to diagnose and reverse.

3

Quality

Testing, observability and shared ownership prevent defects escaping unnoticed.

4

Learning

Blameless review converts failure into changes to the system of work.

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Stakeholders in One Value Stream

Product

- Clarifies value and outcome
- Prioritises by customer impact

Engineering

- Builds in small testable increments
- Owns operability with operations

Operations & Security

- Designs reliability guardrails
- Feeds production learning upstream

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The Core Shift

Optimise the whole system, not a local silo.

A faster development team does not create faster value when review, release or recovery remains constrained.

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TOPIC 01 · R EVIDENCE LAB

Build an Incident-to-Problem Lifecycle Baseline in R

LAB 01

Apply K1 and A1 by managing a supplied incident dataset through detection, logging, investigation, resolution and learning states.

Use R to validate problem records, calculate lifecycle duration, expose ageing work and produce a management summary.

YOU'LL BUILD

A clean lifecycle_summary.csv, ageing_problems.csv and lifecycle_status.png backed by a repeatable R script.

TOOLCHAIN

R 4.x · base R · CSV · PNG evidence

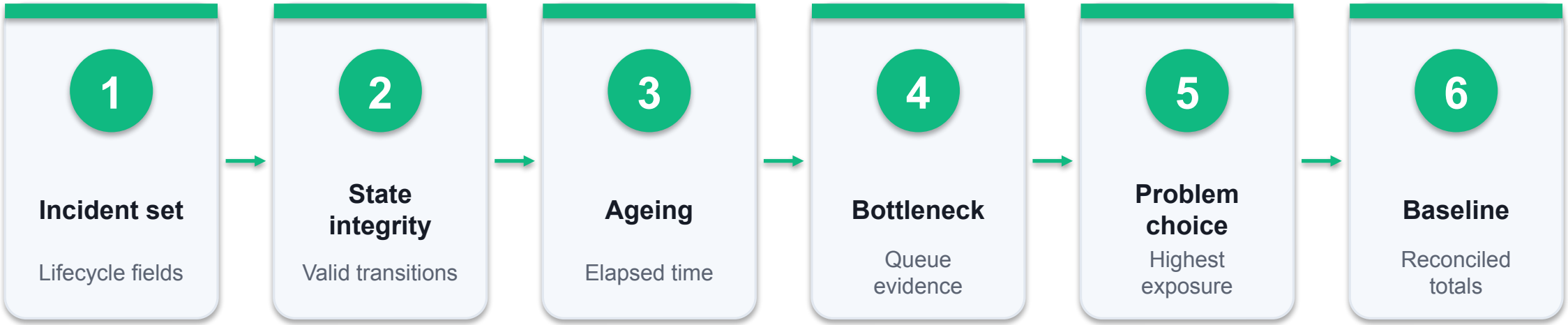
DONE WHEN

The script exits with 'LAB 01 PASS'; all eight records are represented; open problems are ranked by ageing days; status totals reconcile to the source rows.

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LAB 01 · CONCEPTUAL WORKFLOW

Lab 01 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and the lab folder—not on these slides.

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LAB 01 · EVIDENCE CHECK

Build an Incident-to-Problem Lifecycle Baseline in R

✅ Expected result

The script exits with 'LAB 01 PASS'; all eight records are represented; open problems are ranked by ageing days; status totals reconcile to the source rows.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 01 · RETRIEVAL PRACTICE

Recap — Exploring DevOps

1

Explain

People, process and technology optimise one value stream; a tools-only programme is not DevOps.

2

Distinguish

Development, operations, quality and security own flow and reliability together.

3

Apply

Apply K1 and A1 by managing a supplied incident dataset through detection, logging, investigation, resolution and learning states.

4

Evidence

The script exits with 'LAB 01 PASS'; all eight records are represented; open problems are ranked by ageing days; status totals reconcile to the source rows.

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02

TOPIC 02 · LO1 · K2 · A2

Core DevOps Principles

Three Ways · constraints · learning · automated feedback

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BLUEPRINT WEIGHT · 4 / 40 exam questions

Key Ideas — Core DevOps Principles

1

First Way — flow

Visualise work, reduce batch size and remove the system constraint.

2

Second Way — feedback

Build fast signals from production and customers back to the people changing the system.

3

Third Way — learning

Create safety for experimentation, repetition and improvement.

4

Automation supports judgement

Automate repeatable checks; keep people accountable for thresholds, risk and learning.

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TOPIC 02 · K2 · A2

The Three Ways

1

Flow

Left to right value

2

Feedback

Right to left learning

3

Learning

Experiment and improve

SYSTEM VIEW

All three reinforce each other: flow without feedback accelerates mistakes; feedback without learning repeats them.

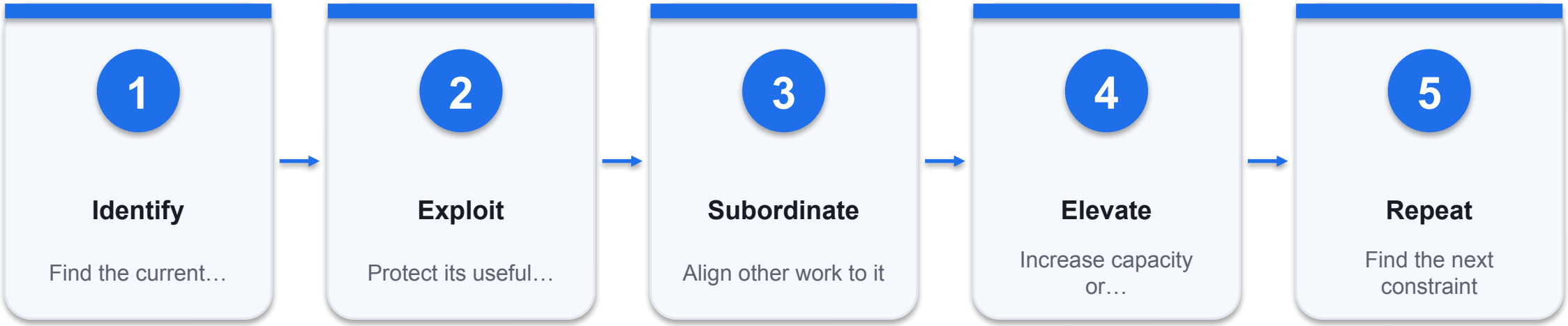
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Traditional Batch vs Continuous Flow

Design choice	Traditional batch	Continuous flow
Change size	Large release package	Small independently releasable change
Integration	Late	At least daily
Feedback	After release window	At every pipeline stage
Recovery	Complex rollback	Fast revert or progressive rollout

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Theory of Constraints Improvement Cycle



CAUTION

Optimising a non-constraint creates local efficiency but not better end-to-end flow.

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Fast Feedback Signals

1

Commit

Static checks and unit tests respond in minutes.

2

Pipeline

Build, integration and security checks create evidence.

3

Production

Logs, metrics, traces and user journeys show real behaviour.

4

Customer

Outcome and support feedback closes the value loop.

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Fail-Safe vs Safe-to-Fail

Question	Fail-safe mindset	Safe-to-fail mindset
Assumption	Failure can be designed out	Failure will occur
Response	Add approval and prediction	Limit impact and recover fast
Practices	Large pre-release gate	Feature flags, canary, rollback
Learning	Find who deviated	Improve conditions and controls

WHEN IT MATTERS Safety culture does not mean accepting reckless change; it means making risk visible, bounded and learnable.

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Automation Principle

Automate the feedback, not the thinking.

A threshold can detect a symptom; people still determine relevance, root cause, priority and the systemic response.

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TOPIC 02 · R EVIDENCE LAB

Detect Actionable Service Signals in R

LAB 02

Apply K2 and A2 by designing an automated detection rule that produces specific, severity-ranked problem signals without excessive noise.

Calculate robust z-scores and consecutive-breach logic from telemetry, then create an actionable alert register.

YOU'LL BUILD

A detected_alerts.csv with severity, evidence and action plus a telemetry_detection.png chart.

TOOLCHAIN

R 4.x · base R · telemetry · automated detection

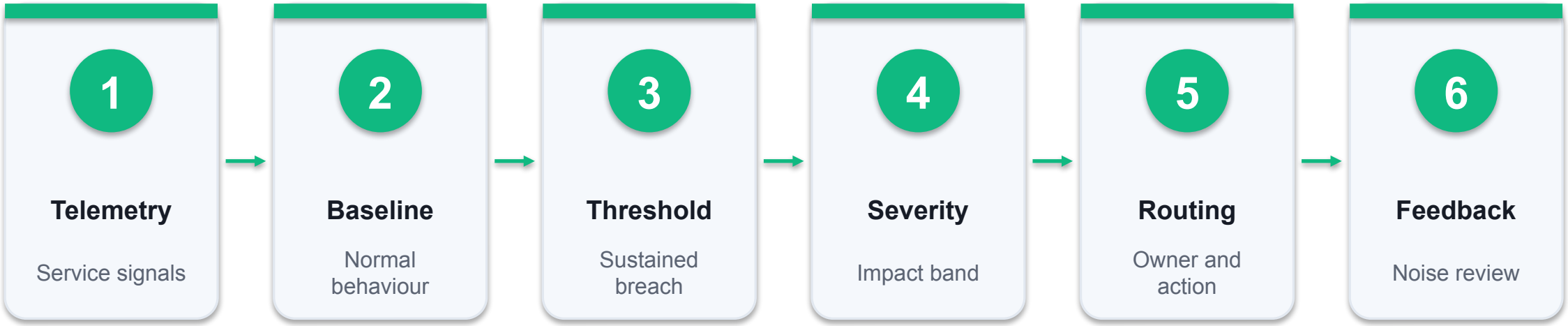
DONE WHEN

The final rule emits exactly two sustained alert windows, no single-point spike alert, and every alert has a non-empty owner and action.

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LAB 02 · CONCEPTUAL WORKFLOW

Lab 02 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and the lab folder—not on these slides.

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LAB 02 · EVIDENCE CHECK

Detect Actionable Service Signals in R

✅ Expected result

The final rule emits exactly two sustained alert windows, no single-point spike alert, and every alert has a non-empty owner and action.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 02 · RETRIEVAL PRACTICE

Recap — Core DevOps Principles

1

Explain

Visualise work, reduce batch size and remove the system constraint.

2

Distinguish

Build fast signals from production and customers back to the people changing the system.

3

Apply

Apply K2 and A2 by designing an automated detection rule that produces specific, severity-ranked problem signals without excessive noise.

4

Evidence

The final rule emits exactly two sustained alert windows, no single-point spike alert, and every alert has a non-empty owner and action.

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03

TOPIC 03 · LO1 · K2 · A2

Key DevOps Practices

CI/CD · testing · SRE · resilience · DevSecOps · ChatOps · Kanban

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BLUEPRINT WEIGHT · 7 / 40 exam questions

Key Ideas — Key DevOps Practices

1

Continuous integration

Frequent integration into a shared mainline, verified by an automated build and tests.

2

Continuous delivery

Every accepted change remains deployable; a business decision may still trigger production release.

3

Continuous deployment

Every accepted change automatically reaches production.

4

Operability by design

SRE, observability, resilience and security are part of the product, not downstream inspections.

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A Modern Delivery Pipeline



CONTROL POINT

Each stage must produce evidence that the next stage can trust.

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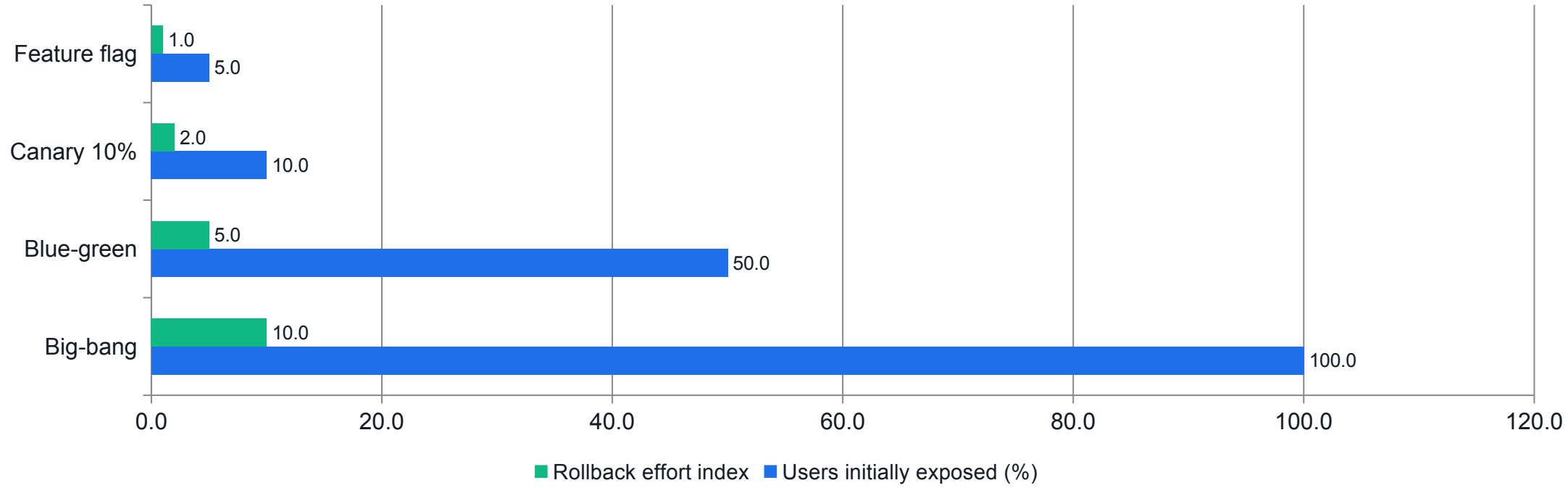
CI, Delivery and Deployment

Practice	End state	Human production decision?
Continuous integration	Tested integrated code	Not applicable
Continuous delivery	Always releasable artifact	Usually yes
Continuous deployment	Accepted change in production	No manual release gate

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TOPIC 03 · K2 · A2

Progressive Delivery Reduces Blast Radius



WHAT THE DATA SHOWS

Progressive strategies reduce initial exposure and make rollback cheaper; they require strong observability.

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Reliability Engineering in Practice

SRE

- Service-level objectives
- Error budgets balance reliability and change

Resilience

- Design for graceful degradation
- Test recovery assumptions

Observability

- Logs, metrics and traces
- Ask new questions without new code

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DevSecOps: Security as Continuous Evidence

1

Shift left

Threat modelling, dependencies and code scanning before merge.

2

Protect pipeline

Signed artifacts, least privilege and isolated secrets.

3

Shift right

Runtime detection, attack simulation and rapid containment.

4

Shared accountability

Security supplies guardrails; product teams own secure outcomes.

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ChatOps and Kanban

Practice	Makes visible	Problem-management value
ChatOps	Conversation, actions and evidence	One shared incident narrative
Kanban	Work, policy and blocked state	Limits work in progress and exposes ageing
Together	Flow plus collaboration	Faster coordinated diagnosis and resolution

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TOPIC 03 · R EVIDENCE LAB

Investigate Recurring Failure Causes with Pareto Analysis

LAB 03

Apply K2 and A2 while preparing evidence for A3: distinguish symptoms from cause candidates and identify recurring categories worth deeper investigation.

Use R to aggregate failure frequency and lost minutes, create a Pareto chart and initialise an evidence-led 5 Whys worksheet.

YOU'LL BUILD

A cause_pareto.csv, pareto_failures.png and five_whys_template.csv.

TOOLCHAIN

R 4.x · base R · Pareto · 5 Whys

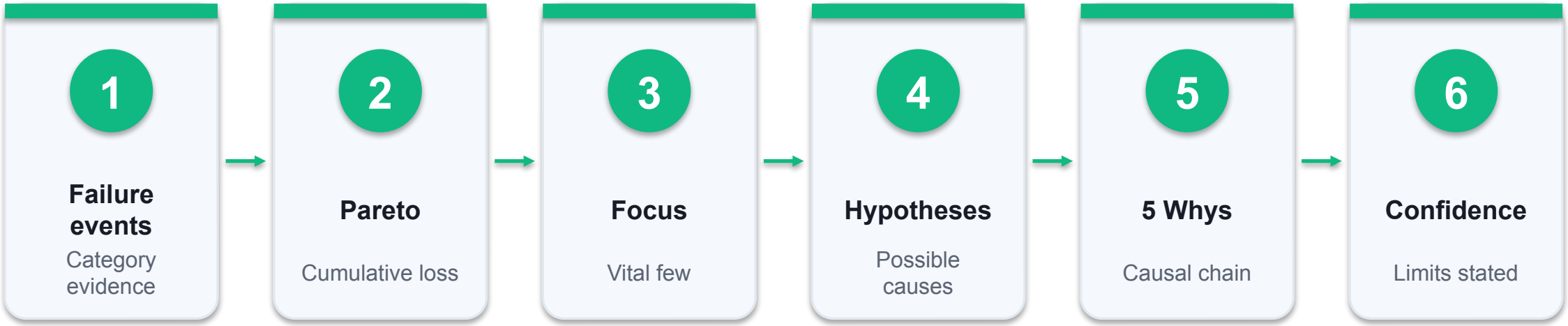
DONE WHEN

LAB 03 PASS is printed; cumulative percentage ends at 100%; the leading category is supported by both count and lost-minute evidence.

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LAB 03 · CONCEPTUAL WORKFLOW

Lab 03 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and the lab folder—not on these slides.

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LAB 03 · EVIDENCE CHECK

Investigate Recurring Failure Causes with Pareto Analysis

✅ Expected result

LAB 03 PASS is printed; cumulative percentage ends at 100%; the leading category is supported by both count and lost-minute evidence.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 03 · RETRIEVAL PRACTICE

Recap — Key DevOps Practices

1

Explain

Frequent integration into a shared mainline, verified by an automated build and tests.

2

Distinguish

Every accepted change remains deployable; a business decision may still trigger production release.

3

Apply

Apply K2 and A2 while preparing evidence for A3: distinguish symptoms from cause candidates and identify recurring categories worth deeper investigation.

4

Evidence

LAB 03 PASS is printed; cumulative percentage ends at 100%; the leading category is supported by both count and lost-minute evidence.

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04

TOPIC 04 · LO2 · K3 · A3

Business and Technology Frameworks

Agile · Lean · ITSM · safety culture · learning organisation

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BLUEPRINT WEIGHT · 7 / 40 exam questions

Key Ideas — Business and Technology Frameworks

1

Agile improves discovery

Short iterations and customer feedback refine what to build.

2

Lean improves flow

Remove delay, overproduction, hand-offs and rework from the value stream.

3

ITSM provides service discipline

Incident, problem and change practices create ownership and traceability.

4

Safety enables truthful evidence

People report weak signals and mistakes when the organisation learns rather than blames.

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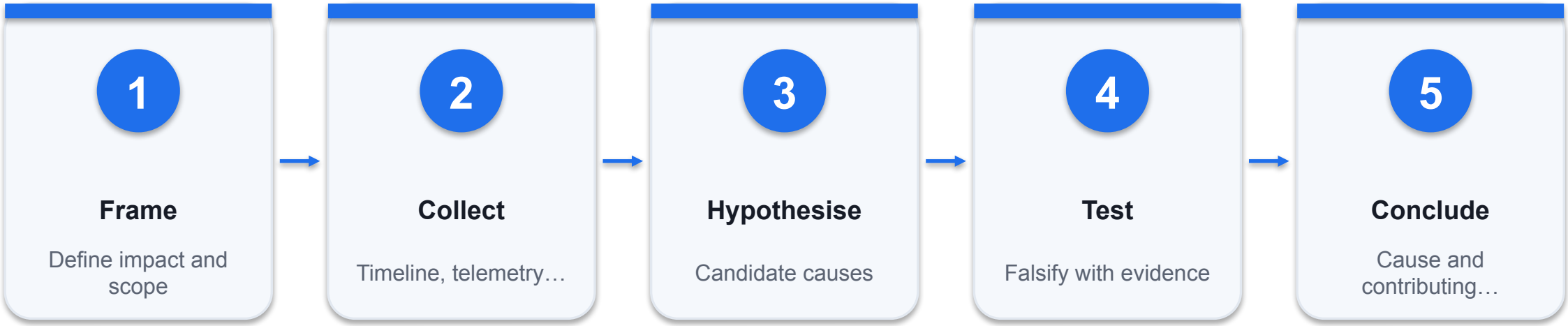
How the Frameworks Complement DevOps

Framework	Primary lens	Contribution
Agile	Adaptive product delivery	Small increments and customer feedback
Lean	End-to-end value flow	Waste reduction and constraint focus
ITSM	Reliable service management	Ownership, records and controlled change
DevOps	Sociotechnical delivery system	Integrates culture, flow, automation and learning

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TOPIC 04 · K3 · A3

Evidence-Led Investigation



A3 STANDARD

Separate observed fact, inference and decision in the problem record.

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Root-Cause Techniques — Choose by Evidence

1

5 Whys

Fast exploration of a causal chain; stop when evidence stops.

2

Fishbone

Organise candidate causes across people, process, technology and environment.

3

Timeline

Reconstruct what changed and when; useful for complex incidents.

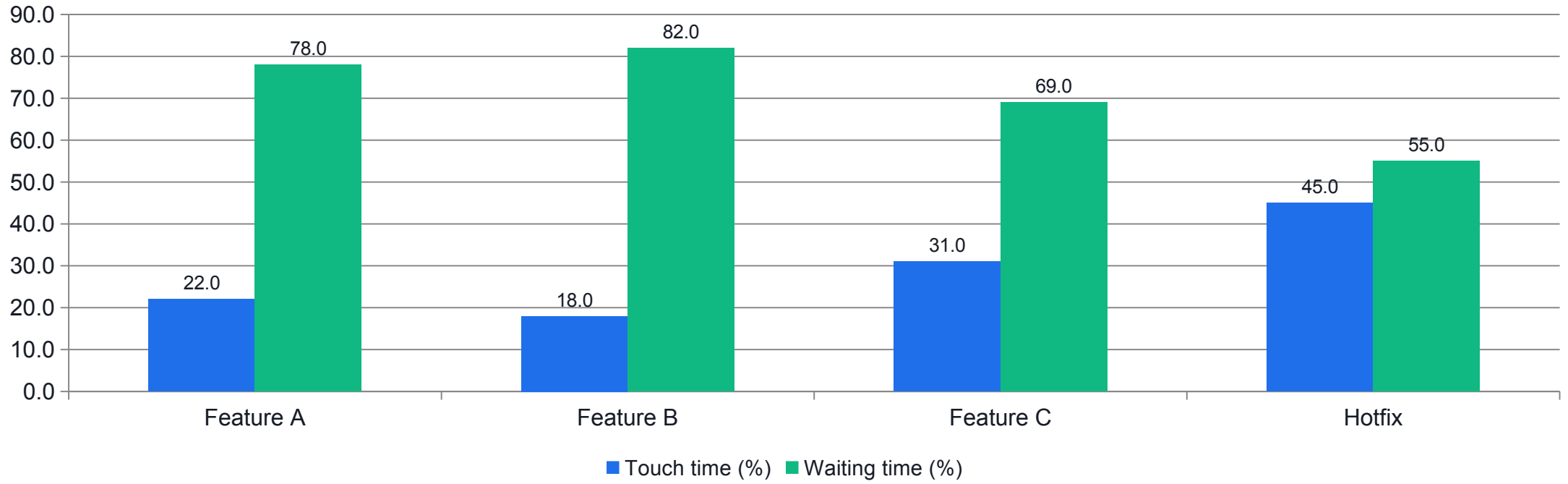
4

Pareto

Identify the few recurring categories that drive most impact.

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Flow Efficiency Example



WHAT THE DATA SHOWS

Most elapsed time is waiting. A Lean investigation asks why work is queued, blocked or handed off.

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Incident vs Problem vs Change

Record	Question	Done when
Incident	How do we restore service?	Impact is controlled
Problem	Why did it happen and recur?	Cause and prevention are managed
Change	How do we alter the service safely?	Evidence shows intended outcome

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Just Culture

Hold people accountable for learning, not for being human.

Examine choices in context, improve the system, and distinguish honest error from reckless behaviour.

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TOPIC 04 · R EVIDENCE LAB

Prioritise Problems by Severity, Frequency and Impact

LAB 04

Apply K3 and A3 by defining evidence, then K4 and A4 by applying a transparent prioritisation rule.

Use R to normalise evidence, calculate a transparent priority score and test how confidence and effort affect sequencing.

YOU'LL BUILD

A ranked problem_priority.csv and priority_matrix.png with documented scoring assumptions.

TOOLCHAIN

R 4.x · base R · prioritisation · impact analysis

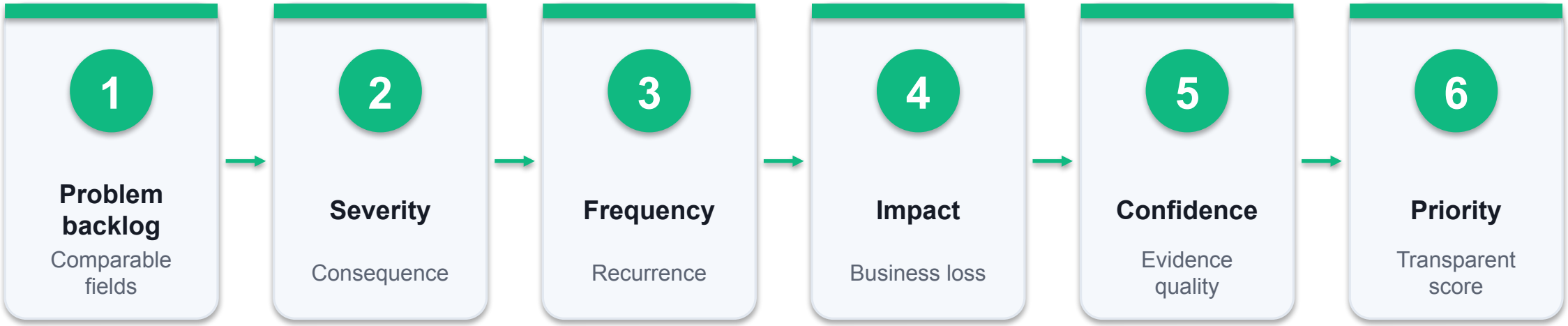
DONE WHEN

Every score is reproducible from the published formula; ranks are unique; the top two recommendations cite impact, evidence confidence and effort.

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LAB 04 · CONCEPTUAL WORKFLOW

Lab 04 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and the lab folder—not on these slides.

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LAB 04 · EVIDENCE CHECK

Prioritise Problems by Severity, Frequency and Impact

✅ Expected result

Every score is reproducible from the published formula; ranks are unique; the top two recommendations cite impact, evidence confidence and effort.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Business and Technology Frameworks

1

Explain

Short iterations and customer feedback refine what to build.

2

Distinguish

Remove delay, overproduction, hand-offs and rework from the value stream.

3

Apply

Apply K3 and A3 by defining evidence, then K4 and A4 by applying a transparent prioritisation rule.

4

Evidence

Every score is reproducible from the published formula; ranks are unique; the top two recommendations cite impact, evidence confidence and effort.

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05

TOPIC 05 · LO2 · K4 · A4

Culture, Behaviours and Operating Models

CALMS · cultural debt · Westrum · maturity · team design

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BLUEPRINT WEIGHT · 6 / 40 exam questions

Key Ideas — Culture, Behaviours and Operating Models

1

Culture is observable behaviour

How information flows, risk is escalated and decisions are made matters more than slogans.

2

Cultural debt compounds

Workarounds, fear and unclear ownership make each future change slower and riskier.

3

Priority policy must be explicit

Severity, frequency, impact and urgency should be scored consistently.

4

Maturity is contextual

Use models to guide conversation and experiments, never as a vanity ranking.

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CALMS as a Transformation Lens

Culture

- Shared ownership
- Psychological safety and trust

Automation & Lean

- Repeatable feedback
- Small batches and reduced waste

Measurement & Sharing

- Outcome evidence
- Learning across boundaries

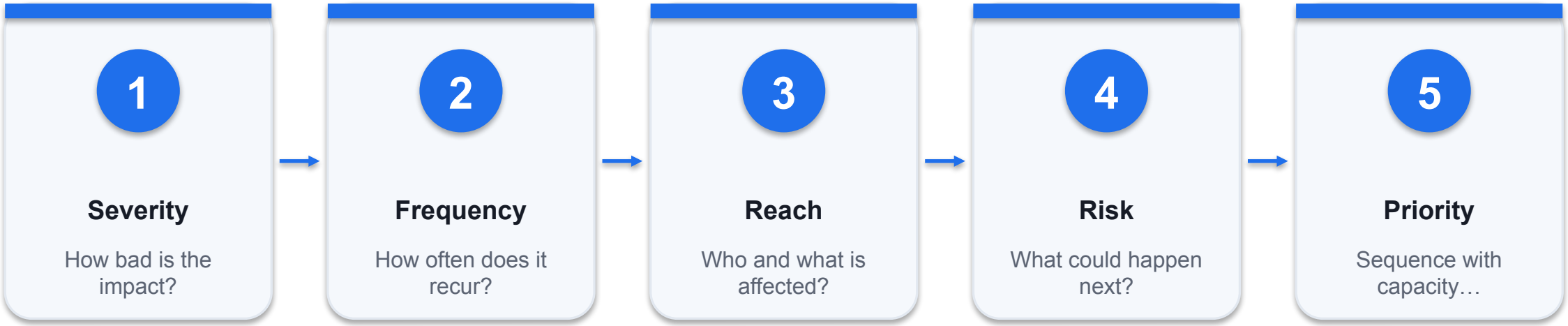
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Westrum Organisational Cultures

Behaviour	Pathological	Bureaucratic	Generative
Information	Hidden	Channels enforced	Actively sought
Failure	Scapegoat	Local justice	Inquiry and learning
Novelty	Crushed	Creates problems	Implemented
Responsibility	Avoided	Compartmentalised	Shared

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From Signal to Priority



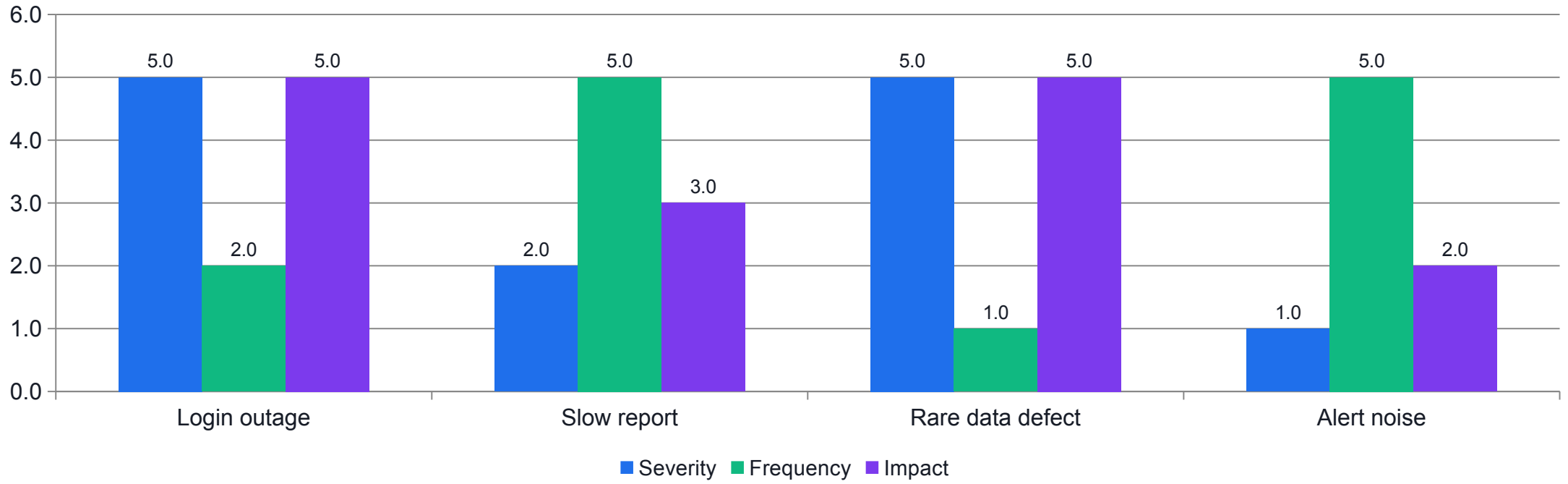
A4 OUTPUT

Publish the scoring rule so two teams reach comparable decisions from the same evidence.

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TOPIC 05 · K4 · A4

Priority Is Not Severity Alone



WHAT THE DATA SHOWS

A severe rare issue and a lower-severity recurring issue can both deserve attention for different reasons.

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Team Operating Models

1

Product team

Owns customer outcome and service operation end to end.

2

Platform team

Provides a paved road that reduces cognitive load.

3

Enabling team

Builds capability temporarily, then moves on.

4

Complicated subsystem

Owns specialist components requiring deep expertise.

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Maturity Principle

Improve the next constraint; do not perform maturity theatre.

A useful maturity model reveals the next experiment and evidence of progress—not a trophy score.

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TOPIC 05 · R EVIDENCE LAB

Diagnose Pipeline Flow and Cultural Debt

LAB 05

Apply K4 and A4 by combining stage performance with visible queue and rework evidence to recommend a prioritised improvement.

Use R to calculate stage wait ratio, failure rate and rework burden, then identify the actual constraint without blaming a team.

YOU'LL BUILD

A pipeline_stage_summary.csv, pipeline_flow.png and a generative-culture recommendation.

TOOLCHAIN

R 4.x · base R · flow metrics · culture evidence

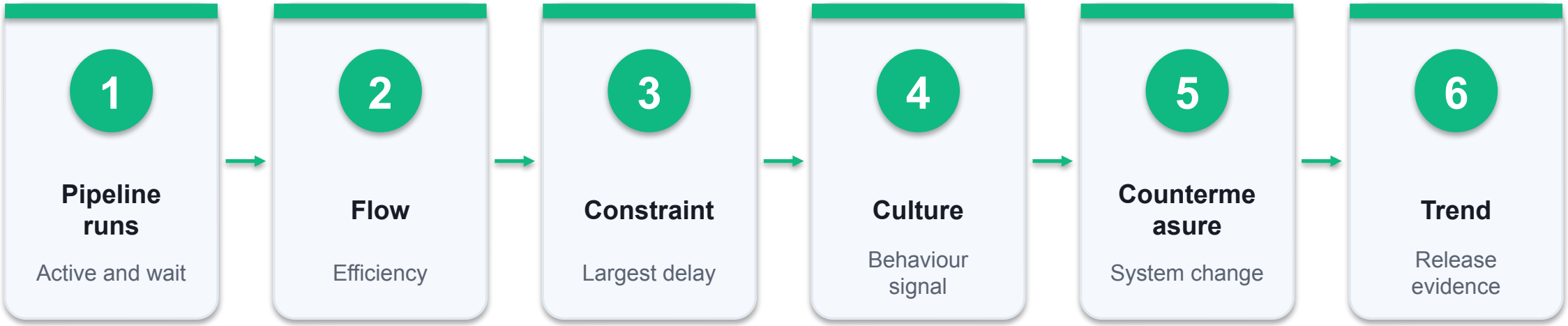
DONE WHEN

The identified constraint matches the maximum wait ratio; the recommendation changes a system condition and names an outcome measure.

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LAB 05 · CONCEPTUAL WORKFLOW

Lab 05 Evidence Model



PROCEDURE BOUNDARY

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LAB 05 · EVIDENCE CHECK

Diagnose Pipeline Flow and Cultural Debt

✅ Expected result

The identified constraint matches the maximum wait ratio; the recommendation changes a system condition and names an outcome measure.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Culture, Behaviours and Operating Models

1

Explain

How information flows, risk is escalated and decisions are made matters more than slogans.

2

Distinguish

Workarounds, fear and unclear ownership make each future change slower and riskier.

3

Apply

Apply K4 and A4 by combining stage performance with visible queue and rework evidence to recommend a prioritised improvement.

4

Evidence

The identified constraint matches the maximum wait ratio; the recommendation changes a system condition and names an outcome measure.

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06

TOPIC 06 · LO2 · K4 · A4

Automation and DevOps Toolchains

CI/CD · cloud · containers · microservices · IaC · AIOps · platform engineering

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BLUEPRINT WEIGHT · 5 / 40 exam questions

Key Ideas — Automation and DevOps Toolchains

1

Design the value stream first

A toolchain should serve a clear flow and evidence model, not automate a broken process.

2

Everything important is versioned

Code, configuration, infrastructure, policy and documentation become reviewable and reproducible.

3

Loose coupling enables flow

Services and teams can test, deploy, scale and fail independently.

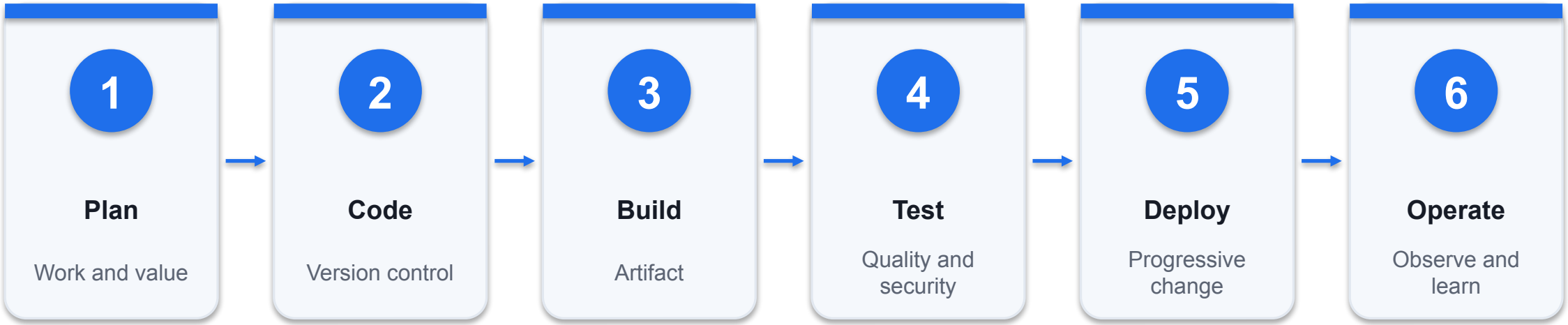
4

Platforms create safe self-service

A paved road standardises guardrails while preserving team autonomy.

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Reference DevOps Toolchain



ARCHITECTURE RULE

Integrate evidence and ownership between tools; avoid a collection of disconnected products.

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Architecture Choices

Choice	Benefit	Cost / risk
Cloud	Elastic self-service resources	Cost, identity and provider governance
Containers	Portable immutable packaging	Image and runtime security
Microservices	Independent change and scale	Distributed-system complexity
IaC	Repeatable reviewed infrastructure	State, drift and secret handling

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Automation Selection Test

1

Repeatable

The task occurs often enough to justify automation.

2

Deterministic

Inputs, rules and acceptable outputs can be specified.

3

Observable

Success and failure produce usable evidence.

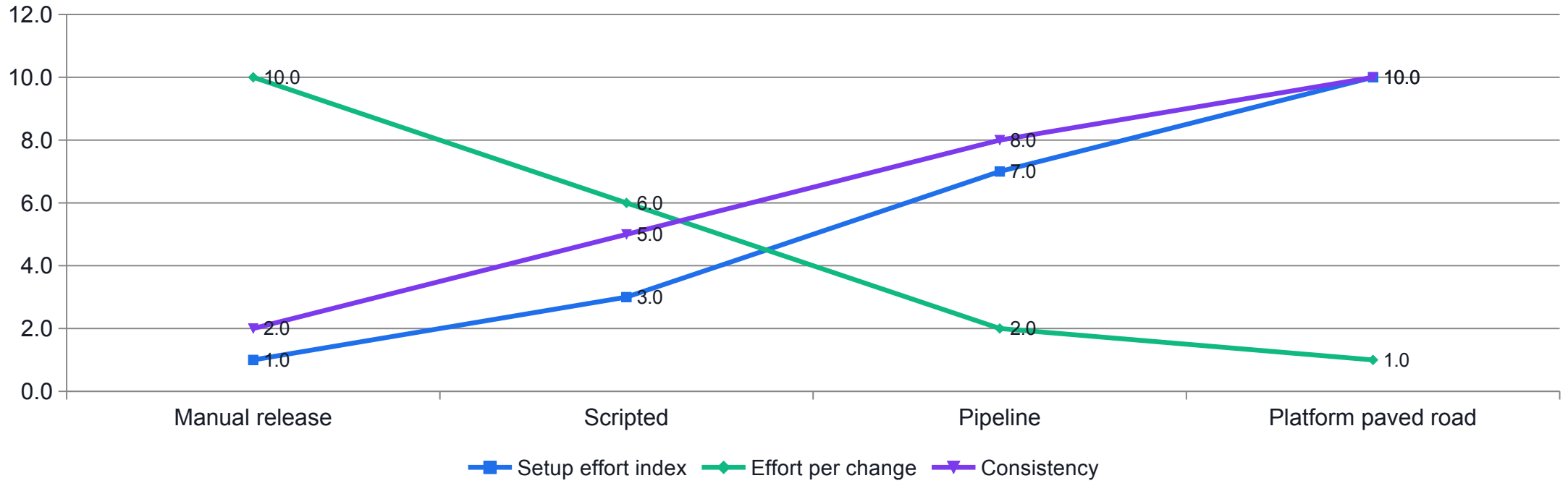
4

Recoverable

A failed automation can stop, retry or roll back safely.

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Automation Economics



WHAT THE DATA SHOWS

More capable automation costs more initially but reduces effort and variation across repeated changes.

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AIOps and Generative AI — Useful Roles

Detect

- Cluster related signals
- Identify unusual patterns

Assist

- Summarise evidence
- Suggest hypotheses for human review

Govern

- Track confidence and provenance
- Require approval for high-impact action

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Toolchain vs Platform

Dimension	Toolchain	Internal platform
Focus	Connected lifecycle tools	Self-service developer experience
User action	Navigate each tool	Use a paved workflow or portal
Governance	Policies per tool	Guardrails embedded in the path
Success	Integrated evidence	Reduced cognitive load and lead time

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TOPIC 06 · R EVIDENCE LAB

Balance Reliability, Security and Delivery Guardrails

LAB 06

Apply K4 and A4 by categorising reliability and security risks, then selecting proportional automated guardrails.

Use R to calculate error-budget burn and security risk, then recommend automate, warn or block policies.

YOU'LL BUILD

A guardrail_decisions.csv, error_budget.png and documented trade-off decision.

TOOLCHAIN

R 4.x · base R · SLO · DevSecOps · policy evidence

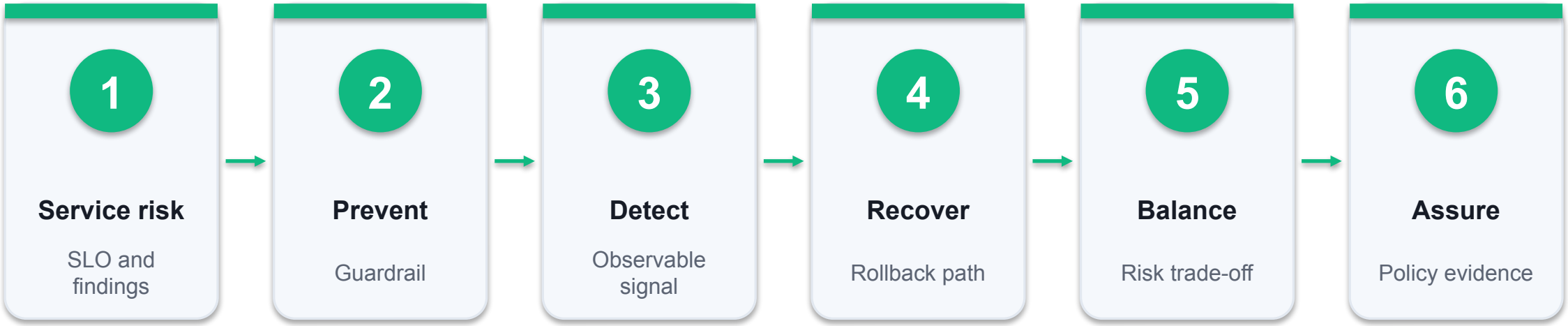
DONE WHEN

All policies are derived from published thresholds; at least one service is blocked, one warns and one proceeds automatically.

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LAB 06 · CONCEPTUAL WORKFLOW

Lab 06 Evidence Model



PROCEDURE BOUNDARY

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LAB 06 · EVIDENCE CHECK

Balance Reliability, Security and Delivery Guardrails

✅ Expected result

All policies are derived from published thresholds; at least one service is blocked, one warns and one proceeds automatically.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Automation and DevOps Toolchains

1

Explain

A toolchain should serve a clear flow and evidence model, not automate a broken process.

2

Distinguish

Code, configuration, infrastructure, policy and documentation become reviewable and reproducible.

3

Apply

Apply K4 and A4 by categorising reliability and security risks, then selecting proportional automated guardrails.

4

Evidence

All policies are derived from published thresholds; at least one service is blocked, one warns and one proceeds automatically.

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TOPIC 07 · LO3 · K5 · A6

Measurement, Metrics and Reporting

DORA · reliability · flow · business value · problem records

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BLUEPRINT WEIGHT · 2 / 40 exam questions

Key Ideas — Measurement, Metrics and Reporting

1

Measure outcomes and system health

A balanced view covers throughput, instability, reliability, quality and business value.

2

Trends beat targets

Use metrics to learn over time, not to rank individuals or encourage gaming.

3

Context matters

Compare one service with itself and record changes in architecture, demand and risk.

4

Documentation is operational telemetry

Problem records show whether causes, actions and verification are actually closed.

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Current DORA Software Delivery Metrics

Throughput

- Change lead time
- Deployment frequency
- Failed deployment recovery time

Instability

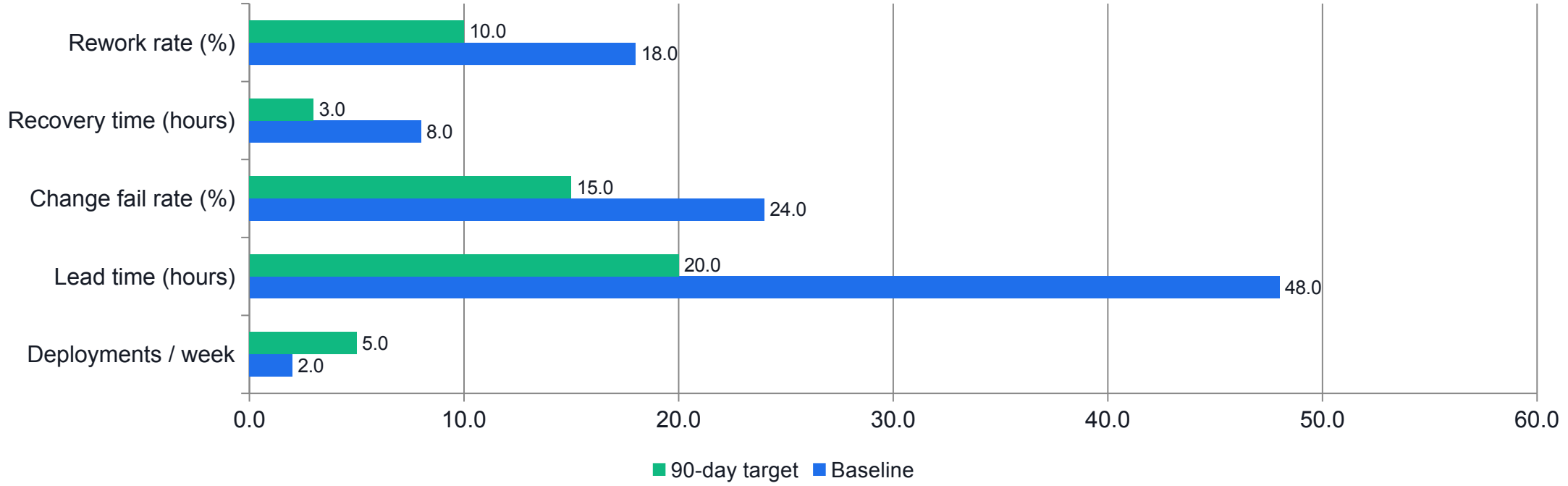
- Change fail rate
- Deployment rework rate

Context

- Use per service
- Pair with reliability and business outcomes

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Baseline and Target — Example Service



WHAT THE DATA SHOWS

The chart mixes units deliberately for a leadership conversation; operational dashboards should separate scales and retain definitions.

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Metric, KPI, SLI, SLO and SLA

Term	Meaning	Example
Metric	Any quantitative measure	Request latency
KPI	Measure tied to a goal	Checkout completion rate
SLI	Observed service measure	99.93% successful requests
SLO	Internal reliability target	99.9% success over 30 days
SLA	External commitment	Credit if monthly availability falls below contract

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Measurement-to-Improvement Loop



ANTI-PATTERN

A dashboard without a decision cadence is decoration.

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Problem Record — Minimum Evidence

1. **Identity**
   Unique ID, service, owner, status and timestamps.

2. **Impact**
   Users, duration, severity and business consequence.

3. **Analysis**
   Timeline, evidence, cause and contributing conditions.

4. **Action**
   Recommendation, accountable owner, due date and verification.

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Healthy vs Harmful Metrics

Healthy use	Harmful use
Team learns from trends	Individuals are ranked by output
Definitions are explicit	Numbers are compared across unlike services
Qualitative context is retained	One target becomes the whole goal
Measures drive experiments	Measures drive blame or gaming

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TOPIC 07 · R EVIDENCE LAB

Build a DORA and Problem-Management Evidence Dashboard

LAB 07

Apply K5 and A6 by calculating documented delivery metrics and linking them to tracked problem-management actions.

Use R to calculate current DORA throughput and instability metrics, connect them to repeat problems and produce a traceable dashboard.

YOU'LL BUILD

A dora_metrics.csv, problem_actions.csv and dora_dashboard.png with metric definitions.

TOOLCHAIN

R 4.x · base R · DORA · problem records · reporting

DONE WHEN

Five DORA measures are present; the deployment counts reconcile; every open action has an owner and due date; dashboard values match the CSV.

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Lab 07 Evidence Model



PROCEDURE BOUNDARY

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LAB 07 · EVIDENCE CHECK

Build a DORA and Problem-Management Evidence Dashboard

✅ Expected result

Five DORA measures are present; the deployment counts reconcile; every open action has an owner and due date; dashboard values match the CSV.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Measurement, Metrics and Reporting

1

Explain

A balanced view covers throughput, instability, reliability, quality and business value.

2

Distinguish

Use metrics to learn over time, not to rank individuals or encourage gaming.

3

Apply

Apply K5 and A6 by calculating documented delivery metrics and linking them to tracked problem-management actions.

4

Evidence

Five DORA measures are present; the deployment counts reconcile; every open action has an owner and due date; dashboard values match the CSV.

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08

TOPIC 08 · LO3 · K5 · A5

Sharing, Shadowing and Evolving

Enterprise adoption · leadership · roles · communities · critical success factors

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BLUEPRINT WEIGHT · 4 / 40 exam questions

Key Ideas — Sharing, Shadowing and Evolving

1

Adoption is an evolving product

Start with a value stream, learn, improve the platform and expand through evidence.

2

Leaders design conditions

Align incentives, fund products, remove systemic constraints and model learning behaviour.

3

Sharing scales learning

Communities, pairing, shadowing and reusable patterns prevent isolated success.

4

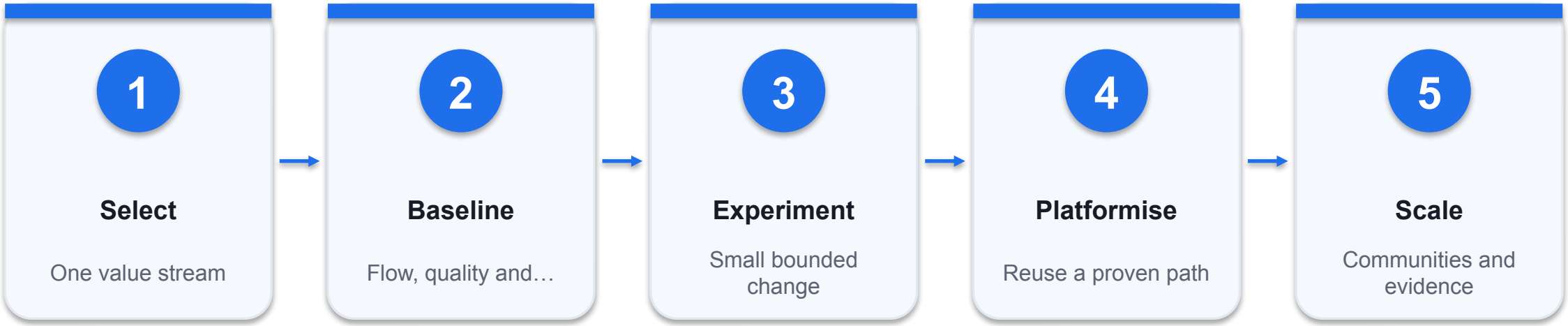
Recommendations need verification

State cause, action, owner, due date, risk and the signal that proves recurrence fell.

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TOPIC 08 · K5 · A5

A Safe Enterprise Adoption Path



CRITICAL SUCCESS FACTOR

Scale learning and enabling capabilities before mandating standard tools everywhere.

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Leadership Responsibilities

Direction

- Connect DevOps to business outcomes
- Make trade-offs explicit

System

- Fund products and platforms
- Remove conflicting incentives

Learning

- Model blameless inquiry
- Protect time for improvement

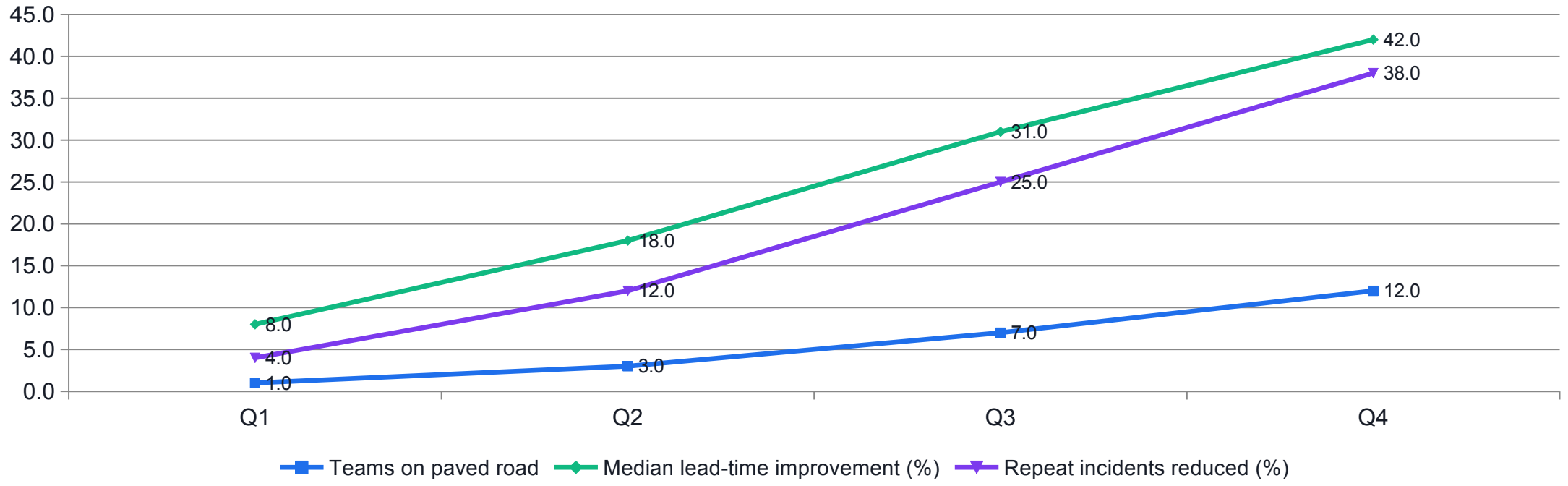
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Ways to Spread Capability

Mechanism	Best use	Watch for
Community of practice	Shared patterns and learning	Meeting without ownership
Pairing / shadowing	Tacit skill transfer	Dependency on one expert
Enablement team	Accelerated adoption	Permanent ticket queue
Internal platform	Safe repeatable self-service	Platform built without user research

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Adoption Evidence Over Four Quarters



WHAT THE DATA SHOWS

Scale is credible when adoption and outcome evidence rise together—not when tool licences alone increase.

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Critical Success Factors

1. **Outcome**
   A visible business and service problem worth solving.

2. **Ownership**
   Cross-functional team owns build, run and learn.

3. **Architecture**
   Small independently changeable components where practical.

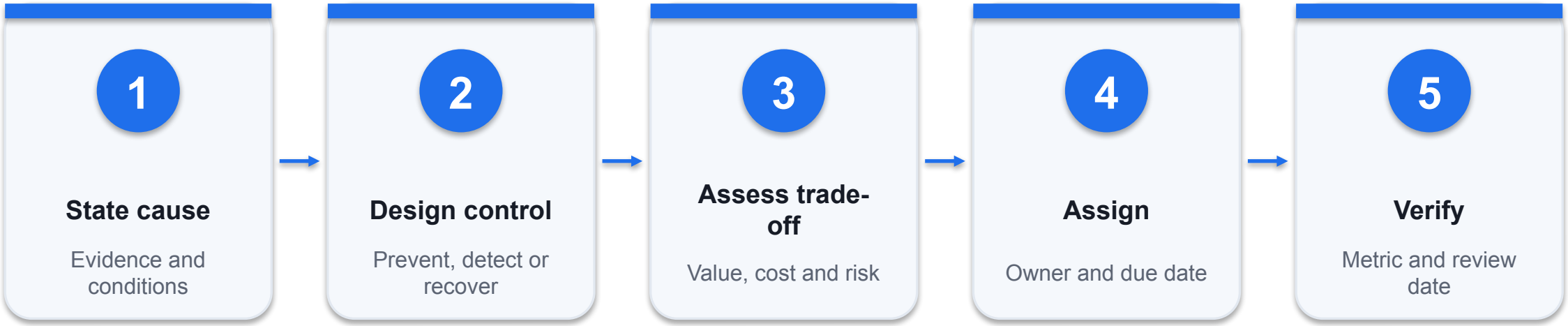
4. **Feedback**
   Trusted telemetry, customer signals and regular review.

5. **Platform**
   Low-friction compliant paths for common work.

6. **Leadership**
   Funding, incentives and behaviour support long-term change.

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From Root Cause to Verified Recommendation



A5 OUTPUT

A recommendation is incomplete until its success criterion and follow-up are recorded.

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Capstone — Produce a Verified DevOps Problem Recommendation

LAB 08

Apply A1–A6 in one connected case: manage the lifecycle, analyse root cause, prioritise, recommend controls and create monitorable documentation.

Use R to combine incident, change, telemetry and action evidence into a concise problem record and recommendations register.

YOU'LL BUILD

A problem_record.md and recommendations.csv that can be audited from detection through verification.

TOOLCHAIN

R 4.x · base R · evidence synthesis · problem record

DONE WHEN

All referenced IDs reconcile; the record covers detection through learning; three recommendations include owner, due date and verification; unsupported causal claims are labelled as hypotheses.

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LAB 08 · CONCEPTUAL WORKFLOW

Lab 08 Evidence Model



PROCEDURE BOUNDARY

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LAB 08 · EVIDENCE CHECK

Capstone — Produce a Verified DevOps Problem Recommendation

✅ Expected result

All referenced IDs reconcile; the record covers detection through learning; three recommendations include owner, due date and verification; unsupported causal claims are labelled as hypotheses.

IF IT DOESN'T WORK

Input mismatch

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Validation stop

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Evidence differs

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Recap — Sharing, Shadowing and Evolving

1

Explain

Start with a value stream, learn, improve the platform and expand through evidence.

2

Distinguish

Align incentives, fund products, remove systemic constraints and model learning behaviour.

3

Apply

Apply A1–A6 in one connected case: manage the lifecycle, analyse root cause, prioritise, recommend controls and create monitorable documentation.

4

Evidence

All referenced IDs reconcile; the record covers detection through learning; three recommendations include owner, due date and verification; unsupported causal claims are labelled as hypotheses.

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END

WRAP-UP

Synthesis, Assessment & Next Steps

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LO1 → LO2 → LO3

The Connected DevOps Problem-Management System



CAPSTONE

Lab 08 and the Case Study require one traceable narrative across the complete lifecycle.

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LEARNING OUTCOMES

What You Can Now Do

1

LO1

Manage a problem lifecycle and introduce automated detection with DevOps practices.

2

LO2

Investigate evidence, analyse root causes and apply a transparent priority policy.

3

LO3

Recommend preventive, detective and recovery controls and monitor the problem record.

4

Professional practice

Explain trade-offs across flow, reliability, security, culture and business value.

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CONTINUE YOUR LEARNING

Recommended Courses

1. **1**
   Designing and Implementing Microsoft DevOps Solutions (AZ-400)
2. **2**
   Application Integration with Docker and Kubernetes
3. **3**
   DevOps Engineering on AWS
4. **4**
   Mastering Agile Project Management for IT Projects
5. **5**
   Kubernetes and Cloud Native Associate (KCNA) Training

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READY

Assessment — Final Check

1

WA-SAQ

Five open-ended knowledge questions · K1–K5 · 60 minutes.

2

Case Study

Three connected ability tasks · A1–A6 · 60 minutes.

3

Digital attendance

Complete the SSG assessment attendance and TRAQOM steps.

4

Submission

Upload candidate papers through LMS-TMS; keys remain trainer-only.

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ON ASSESSMENT DAY

Assessment Flow



APPEAL

If you disagree with an assessment decision, use the documented reassessment and appeal process.

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Digital Attendance (Mandatory)

1

AM

Scan and submit at the start of the day.

2

PM

Scan and submit after lunch.

3

Assessment

Scan before starting the final assessment.

4

Confirm

Do not close the page until submission succeeds.

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DEVOPS FOUNDATION

Thank You!

Keep flow visible, investigate with evidence, and turn every problem into verified learning.

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